

Rolling and Denoising: Rethinking Dynamic Modal Fusion for Multi-Modal Object Re-Identification

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 <https://github.com/lsh-ahu/RoDI>

Abstract

Multi-modal object Re-Identification (ReID) aims to effectively fuse heterogeneous identity representations from various modalities. However, existing methods lack reliable measures of fusion learning quality, leading to two major limitations: 1) fusion imbalance, where high-quality dominant modalities are preferentially emphasized, suppressing the contribution of weaker ones, and 2) fusion conflict, where low-quality modalities with noise are hardly integrated and conflict with easily fused samples during training. To address these limitations, we propose a novel framework, **Rolling and Denoising (RoDI)**, which provides a comprehensive evaluation and optimization of multi-modal fusion learning. Specifically, we first propose an Evidence-aware Dynamic Fusion Learning (EDFL) module that quantifies imbalance and conflict via subjective opinions during modality fusion. Then, we propose a Modality Synergistic Rolling (MSR) module that estimates an expected balanced anchor from the imbalance mass and breaks the fusion imbalance by rotating the fusion queries. Finally, a Local Modality Denoising (LMD) module is proposed to identify critical local patch tokens based on belief mass and reconstruct high-quality pseudo-modal features through diffusion-based denoising, thereby alleviating fusion conflicts. Extensive experiments on four public datasets demonstrate the effectiveness of our method, showing significant improvements over existing state-of-the-art methods.

1. Introduction

Multi-modal object Re-Identification (ReID) aims to achieve robust retrieval of target queries by introducing heterogeneous modalities [49–51], *i.e.*, Visible Light (RGB), Near Infrared (NIR), and Thermal Infrared (TIR). It overcomes the challenges of target identification [14, 21, 42]

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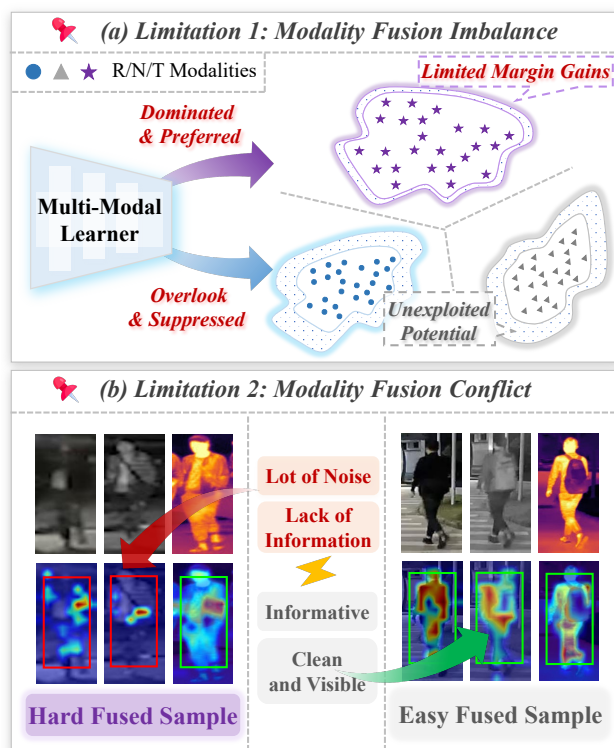


Figure 1. (a) Dominant modalities bias the learner, restricting its capacity to mine identity features from weaker ones and resulting in limited marginal gains. (b) Low-quality modalities contain excessive noise and limited identity information, hindering fusion and negatively affecting clean and easily fused samples.

under adverse conditions such as appearance blur, illumination pollution, color degradation, and weather interference [3, 20, 28]. Despite these advantages, modal heterogeneity and noisy low-quality inputs still hinder effective feature fusion across spectral domains. Recently, existing studies have explored fusion learning from diverse perspectives, including fusion decoupling [32–34, 48], frequency enhancement [41, 45], structural modeling [30, 31], and ap-

pearance semantics [17–19, 35]. Although these methods differ in how they obtain fused representations, their fusion paradigm are largely similar. They typically extract effective local features [35, 45, 48] within each modality and perform inter-modal interaction [17, 30, 32, 34, 48] to produce discriminative fused identity features. Consequently, the fusion performance hinges on the quality of intra-modal feature selection and the reliability of inter-modal interaction.

By dynamically fusing multiple spectral, existing multi-modal methods construct richer and more robust identity representations. However, achieving ideal and complete fusion still faces two major limitations. **Limitation 1: Fusion Imbalance. The model tends to favor one dominant modality while lacking the ability to extract discriminative cues from weaker ones.** In practice, this bias leads to inefficient fusion learning [2, 23, 24, 39], as attention becomes overly concentrated on the dominant modality with limited marginal improvement, as shown in Fig. 1 (a). Recent findings [23, 39] suggest that repeated optimization under such bias exacerbates attention accumulation, making the model insensitive to data quality and less responsive to weaker modalities.

Another key factor that affects fusion performance is **Limitation 2: Fusion Conflict. Low-quality modalities contain limited informative features, making it difficult to provide sufficient complementary cues during multi-modal fusion.** As illustrated in Fig. 1 (b), existing local feature selection methods [35, 45] jointly extract local features from different modalities without accounting for selection quality. Compared with easily fused samples, these conflicting samples introduce a negative impact [8, 40] that degrades the efficiency of fusion learning. Although several studies [8, 16, 40, 44] attempt to identify conflicts from global modal representations, most of them still lack fine-grained local conflict detection and localized conflict mitigation strategies.

To overcome the above limitations, we propose a novel framework that revisits multi-modal fusion learning from a balanced and conflict-free perspective, called Rolling and Denoising (RoDI). Specifically, we first propose an Evidence-aware Dynamic Fusion Learning (EDFL) framework, which performs fundamental modality fusion based on a vanilla fusion module. By leveraging subjective logic under evidence theory, EDFL obtains subjective opinions during the fusion process, effectively quantifying the local belief mass and fusion uncertainty mass within each modality. Then, we propose a Modality Synergistic Rolling (MSR) module to estimate the expected balanced fusion anchor by deriving imbalance quality from fusion uncertainty. Through collaborative rolling across modalities, MSR refines fusion query features and mitigates imbalance in the fusion process. Finally, we propose a Local Modality

Denoising (LMD) module that identifies key local tokens based on belief mass and diffuses modality features into a Gaussian space, reconstructing high-quality pseudo-modal features through denoising to alleviate fusion conflict. Extensive experiments on four benchmarks demonstrate the effectiveness of our approach. Our contributions are summarized as follows:

- **Rethinking.** We revisit the multi-modal ReID fusion paradigm and propose an Evidence-aware Dynamic Fusion Learning (EDFL) framework that provides a principled measure for balanced and conflict-free fusion.
- **Rolling.** To achieve balanced fusion, we propose a Modality Synergistic Rolling (MSR) module that estimates the expected anchor and rotates fusion queries based on imbalance evidence to break the inherent bias.
- **Denoising.** To address fusion conflict, we propose a Local Modality Denoising (LMD) module that performs diffusion-based reconstruction to generate reliable pseudo-modal features and reduce fusion conflicts.

2. Related Work

2.1. Multi-Modal Object ReID

Multi-modal object ReID aims to achieve robust recognition of target queries by introducing auxiliary modalities [49–51]. To effectively fuse visual features, existing studies have explored fusion learning from diverse perspectives. Wang *et al.* [32] proposes TOP-ReID, which employs a modality permutation and reconstruction mechanism to align multi-modal representations. Zhang *et al.* [48] introduces a prompt-based token selection and fusion strategy to integrate multi-modal features. Wang *et al.* [34] proposes a mixture-of-expert decoupling framework to learn modality-shared and specific features. Zhang *et al.* [45] introduces a frequency-domain enhancement and selection strategy to extract local features while suppressing background interference. Wan *et al.* [30] employs Gaussian graph modeling to capture structural relationships and obtain robust multi-modal representations. Li *et al.* [17] and Wang *et al.* [35] propose multi-modal appearance caption generation pipelines that guide modal feature fusion via text modulation and inversion mechanisms. Despite these advances, existing methods generally follow the line of inter-modal feature interaction and intra-modal local feature mining, without reliable mechanisms for measuring fusion quality. To address these challenges, we revisit the fusion paradigm from the perspectives of modality balance and conflict-free denoising.

2.2. Multi-Modal Learning

Multi-modal learning aims to integrate information [2, 9, 37, 38] from diverse modalities to enhance a model’s perception and reasoning capability. However, recent stud-

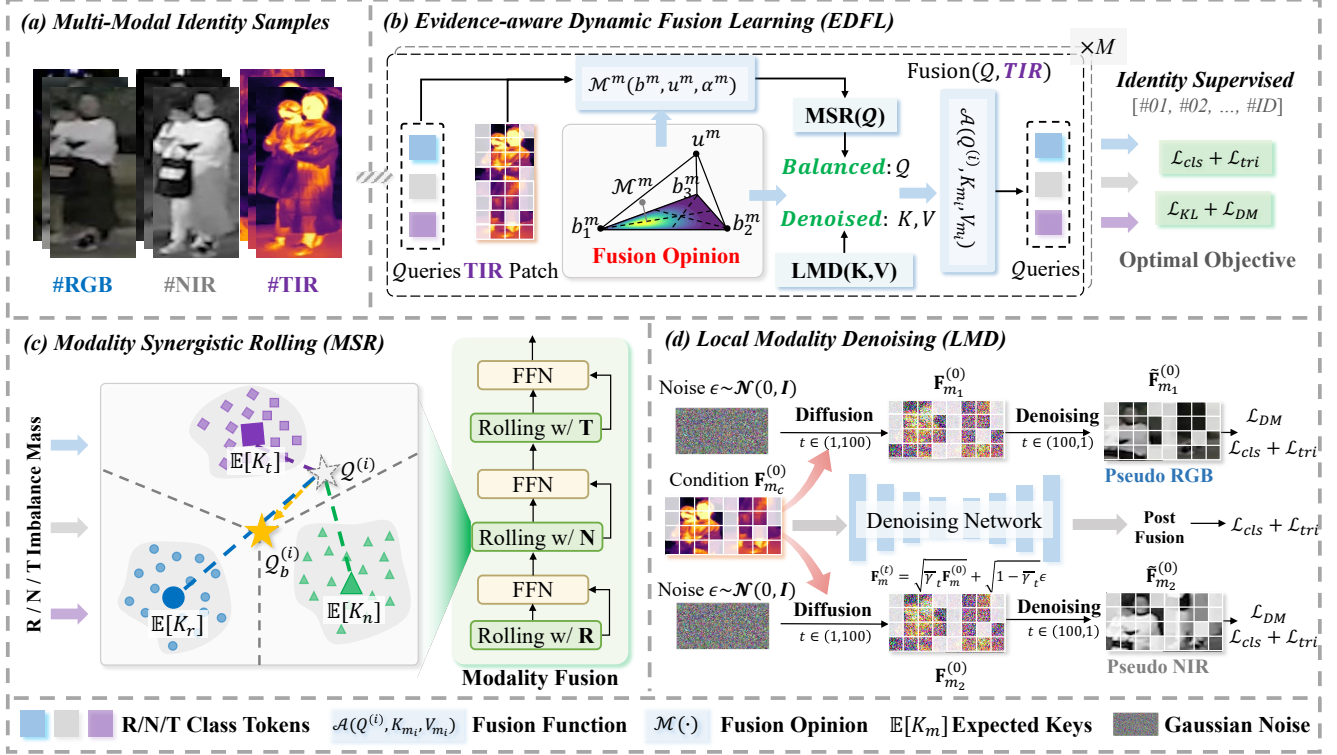


Figure 2. The overview of our proposed method. It mitigates fusion imbalance and fusion conflict by combining three cooperative components: (1) EDFL quantifies belief and uncertainty during fusion and aligns inter-modal evidence. (2) MSR balances modalities by estimating expected anchors and rotating fusion queries. (3) LMD refines local patch tokens via diffusion-based denoising to reconstruct high-quality pseudo-modal features. With the integration of EDFL, MSR, and LMD, RoDI enables interpretable, balanced, and conflict-free multi-modal fusion learning.

ies [2, 23, 24, 37, 39] reveal that multi-modal deep networks often exhibit a greedy learning behavior, where the model over-relies on a dominant modality while neglecting weaker ones, leading to fusion imbalance and performance degradation. To address this issue, Wu *et al.* [39] propose a balance optimization algorithm that operates without additional computational cost, while Chaudhuri *et al.* [2] suppress modality collapse through explicit base redistribution. Hu *et al.* [23] further investigate the influence of uneven modality quality on attention allocation and encoder optimization under dynamic fusion paradigms. Nevertheless, these studies lack reliable measures of fusion quality. On the other hand, real-world data often contain low-quality or conflicting samples, making reliable multi-modal fusion difficult to achieve [5, 16, 46, 47]. Xu *et al.* [40] introduce a reliable conflicting multi-view learning approach that models uncertainty via decision outputs and confidence estimation, whereas Huang *et al.* [8] design a trustworthy unified feature neighborhood dynamic model that integrates local and global neighborhood structures for robust decisions. Although these methods alleviate modality conflicts at the global level, the interference of noisy local features remains underexplored. To this end, we further explore a generative

method that produces high-quality features, mitigating the conflicts caused by low-quality samples.

3. Proposed Method

In this section, we elaborate on the proposed method **RoDI**. Fig. 2 illustrates the dynamic fusion framework that includes Evidence-aware Dynamic Fusion Learning (EDFL) module, Modality Synergistic Rolling (MSR) module, and Local Modality Denoising (LMD) module.

3.1. Preliminary

We first define the key concepts and notations in classic multi-modal fusion learning. Each multi-modal identity sample is denoted as $\mathbf{X} = \{X_m\}$, $m \in \{R, N, T\}$, Y where X_m is composed of several spectral modalities (*i.e.*, visible light R , near-infrared N , and thermal-infrared T) and Y denotes the corresponding identity label. For each modality, a visual encoder $\mathcal{V}(\cdot)$ is applied to extract visual features:

$$\mathbf{F}_m = [\mathbf{f}_m^c; \mathbf{F}_m^p] = \mathcal{V}(\mathbf{X}), \quad (1)$$

where $\mathbf{f}_m^c \in \mathbb{R}^D$ is the *class* token, and $\mathbf{F}_m^p \in \mathbb{R}^{N \times D}$ is the set of patch tokens, with N representing the number of patches and D is the feature dimension.

In general, a vanilla fusion module $\mathcal{F}(\cdot)$ is defined, which leverages a cross-attention mechanism [4] to capture cross-modal dependencies and aggregate them into unified representations. The fusion process is performed sequentially across modalities in a fixed order (*e.g.*, $R \rightarrow N \rightarrow T$), where a single query embedding is progressively fused via $\mathcal{F}(\cdot)$ with each modality. Let $\mathbf{Q}^{(0)} \in \mathbb{R}^{3 \times D}$ denote the initial query derived from concatenated three modality class tokens. At each fusion step i , the query is refined by attending to the patch tokens $\mathbf{F}_{m_i}^p$ of modality m_i using a specific cross-attention module $\mathcal{A}(\cdot)$:

$$\mathbf{Q}^{(i+1)} = \mathcal{A}(\mathbf{Q}^{(i)}, \mathbf{K}_{m_i}, \mathbf{V}_{m_i}), \quad (2)$$

where $\mathbf{K}_{m_i}$ and $\mathbf{V}_{m_i}$ are the key and value projections of the patch tokens from modality m_i , and $\mathbf{Q}^{(i)}$ is the current query state at step i . This progressive fusion continues through all modalities in order. After that, the final query $\mathbf{Q}^{(L)}$ is then passed through a batch normalization layer [10] and a classifier to produce the identity feature $\mathbf{F}_{id} = \text{CLS}(\text{BN}(\mathbf{Q}^{(L)}))$, where L is the length of fusion order.

3.2. Evidence-aware Dynamic Fusion Learning

Most existing multi-modal fusion approaches treat the fusion process as an equal information interaction [32, 33, 48]. They typically lack an effective evaluation of how each modality influences the learning capacity of the fusion module. To solve it, we extend the Evidential Deep Learning (EDL) [26] to enable a fine-grained assessment of each fusion step, as shown in Fig. 2 (b). Based on subjective logic theory [12], we gather subjective opinion $\mathcal{M}(\mathbf{b}, \mathbf{u}, \alpha)$ from data, where $\mathbf{b}$ represents the belief mass, $\mathbf{u}$ denotes the imbalance mass, and α corresponds to the Dirichlet distribution parameters used to model second-order probabilities and imbalance.

The Dirichlet distribution is a probability density function for possible values of the probability mass function $\mathbf{p}$.

$$D(\mathbf{p} | \alpha) = \begin{cases} \frac{1}{B(\alpha)} \prod_{k=1}^K p_k^{\alpha_k - 1} & \text{for } \mathbf{p} \in S_K, \\ 0 & \text{otherwise,} \end{cases} \quad (3)$$

where S_K is the K -dimensional unit simplex,

$$S_K = \left\{ \mathbf{p} \mid \sum_{i=1}^K p_i = 1 \text{ and } 0 \leq p_1, \dots, p_K \leq 1 \right\}, \quad (4)$$

and $B(\alpha)$ is the K -dimensional multinomial beta function [11].

Based on this, we first convert the attention scores into cosine similarity as *evidence* $\mathbf{e}^{(i)} \in \mathbb{R}^{3 \times N}$ in each fusion step i which quantifies the interaction strength between the current query $\mathbf{Q}^{(i)}$ and the patch tokens of modality m_i :

$$\mathbf{e}^{(i)} = \text{Sim}(\mathbf{Q}^{(i)}, \mathbf{K}_{m_i}), \quad (5)$$

where m_i denotes the i -th modality in the fusion sequence.

For the i -th step, the belief mass vector is defined as $\mathbf{b}^{(i)} = \{b_n^{(i)}\}_{n=1}^N$ and the imbalance mass is denoted as $\mathbf{u}^{(i)}$. They are computed as follows:

$$\mathbf{b}^{(i)} = \frac{\mathbf{e}^{(i)}}{S^{(i)}}, \quad \mathbf{u}^{(i)} = \frac{N}{S^{(i)}}, \quad \text{and} \quad \mathbf{b}^{(i)} + \mathbf{u}^{(i)} = 1, \quad (6)$$

where $S^{(i)} = \sum_{n=1}^N \alpha^{(i,n)}$ represents the Dirichlet strength, and $\alpha^{(i,n)} = e^{(i,n)} + 1$ as parameters of the Dirichlet distribution. Based on this evidential perspective, the dynamic fusion process can be observed by the local belief quality and the global imbalance within each modality.

3.3. Modality Rolling for Balanced Fusion

During multi-modal fusion learning, one modality with lower imbalance mass may be greedily favored by the model [2, 9, 39]. As a result, the contribution of the weaker ones is hardly exploitable. To solve it, we propose a Modality Synergistic Rolling (MSR) module that dynamically measures the imbalance gap and adjusts the fusion query $\mathbf{Q}^{(i)}$ to promote balanced interactions among modalities, as shown in Fig. 2 (c).

Specifically, to restore the learning balance, the fusion query $\mathbf{Q}^{(i)}$ is rotated toward an expected anchor $\mathbf{Q}_b^{(i)}$. This rotation encourages the model to compensate for weaker modalities by aligning its attention and reducing the dominance of the biased modality. To determine $\mathbf{Q}_b^{(i)}$, we measure the imbalance mass $\mathbf{u}^{(i)}$ in each mini-batch. A modality with higher imbalance mass is considered less confident and tends to play a weaker role in the fusion process. To quantify this gap, we define a modality weight vector

$$\mathbf{w} = \text{Softmax}\left(\frac{\mathbf{u} - \bar{\mathbf{u}}}{\bar{\mathbf{u}}}\right), \quad (7)$$

where $\mathbf{u} = u^{(1)}, u^{(2)}, u^{(3)}$ denotes the imbalance mass of each modality and $\bar{\mathbf{u}}$ is their mean within the current mini-batch.

After that, the fusion query $\mathbf{Q}^{(i)}$ and each modality key $\mathbf{K}_{m_i}$ are defined as expectation form $\mathbb{E}[\mathbf{Q}^{(i)}]$ and $\mathbb{E}[\mathbf{K}_{m_i}]$, respectively. The re-balanced anchor query $\mathbf{Q}_b^{(i)}$ is then derived as

$$\mathbf{Q}_b^{(i)} = \sum_{i=1}^M w_i \frac{\mathbb{E}[\mathbf{K}_{m_i}]}{\|\mathbb{E}[\mathbf{K}_{m_i}]\|_2} \|\mathbb{E}[\mathbf{Q}^{(i)}]\|_2, \quad (8)$$

where $\|\cdot\|_2$ means L_2 normalization and $M = 3$ is the number of all modalities.

To ensure the observed fusion query $\mathbf{Q}^{(i)}$ is effectively rotated toward the balanced anchor $\mathbf{Q}_b^{(i)}$, a Singular Value

Decomposition [22] (SVD) is applied to compute the corresponding rotation matrix $\mathbf{R}_b^{(i)} \in \mathbb{R}^{3 \times D \times D}$:

$$\mathbf{R}_b^{(i)} = \text{SVD}([\mathbb{E}[\mathbf{Q}^{(i)}], \mathbf{Q}_b^{(i)}]), \quad (9)$$

which satisfies $\mathbf{Q}_b^{(i)} = \mathbb{E}[\mathbf{Q}^{(i)}] \cdot \mathbf{R}_b^{(i)}$. The rebalanced query is thus obtained as $\tilde{\mathbf{Q}}^{(i)} = \mathbf{Q}^{(i)} \cdot \mathbf{R}_b^{(i)}$.

To enhance evidence consistency across modalities, we adopt the Kullback-Leibler (KL) divergence [1] to align the Dirichlet distributions of the attention. The loss term is defined as

$$\mathcal{L}_{KL} = \frac{1}{L} \sum_{i=1}^L \text{KL}(\text{Dir}(\boldsymbol{\alpha}^{(i)}) \parallel \text{Dir}(\bar{\boldsymbol{\alpha}})). \quad (10)$$

where $\bar{\boldsymbol{\alpha}} = \frac{1}{L} \sum_{i=1}^L \boldsymbol{\alpha}^{(i)}$, and $\text{Dir}(\cdot)$ represents the Dirichlet distribution with parameter $\boldsymbol{\alpha}$.

By jointly considering inter-modal learning bias, the fusion query is shaken off the over-reliance on dominant modalities and promotes a fairer, more balanced fusion process.

3.4. Modality Denoising for Conflict-Free Fusion

Another key challenge in dynamic fusion learning lies in conflicting modality pairs, which manifest as noisy or low-belief mass patches in certain inputs. Unlike clean samples, these noisy and hardly fused cases negatively impact the optimization of the fusion module. As shown in Fig. 2 (d), a Local Modality Denoising (LMD) is proposed to identify critical patch tokens that cause conflicts, and a conditional denoise mechanism is applied to transfer features across modalities.

Specifically, we first fuse the subjective opinions $\mathcal{M}$ in each fusion step to obtain the union opinions $\tilde{\mathcal{M}}$ as follows:

$$\tilde{\mathbf{b}}^{(L)} = \frac{\sum_{i=1}^L \mathbf{b}^{(i)} \cdot \mathbf{u}^{(i)}}{\sum_{i=1}^L \mathbf{u}^{(i)}}, \tilde{\mathbf{u}}^{(L)} = \frac{L \cdot \prod_{i=1}^L \mathbf{u}^{(i)}}{\sum_{i=1}^L \mathbf{u}^{(i)}}, \quad (11)$$

where $\tilde{\mathbf{b}}^{(L)} \in \mathbb{R}^{3 \times N}$ is the fused belief mass, and $\tilde{\mathbf{u}}^{(L)} \in \mathbb{R}^3$ denotes the fused imbalance mass.

The fused opinions among fusion queries are combined in the same manner as

$$\mathbf{B} = \frac{\sum_{m=1}^M \tilde{\mathbf{b}}^{(L,m)} \cdot \tilde{\mathbf{u}}^{(L,m)}}{\sum_{m=1}^M \tilde{\mathbf{u}}^{(L,m)}}, \quad (12)$$

where $M = 3$ is the number of all fusion queries. Based on this, the TOP- K local patch tokens with the highest belief mass are selected as the de-conflicted features $\tilde{\mathbf{F}}_m^p = \text{TOP-}K(\mathbf{F}_m^p, \mathbf{B})$.

Then, following the Denoising Diffusion Probabilistic Model (DDPM) [7], the modality feature $\mathbf{F}_m = [\mathbf{f}_m^c; \tilde{\mathbf{F}}_m^p]$

is assumed to follow a specific distribution as $\mathbf{F}_m^{(0)} \sim q(\mathbf{F}_m^{(0)})$. The cross-modal de-conflicting process is treated as a conditional transformation between Gaussian distributions, given by $\mathbf{F}_m^{(0)} \sim q(\mathbf{F}_m^{(0)}; \mathbf{F}_{m_c}^{(0)})$, where c denotes the conditional modalities.

The forward diffusion process adds Gaussian noise $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ to $\mathbf{F}_m^{(0)}$ step by step for T iterations under a variance schedule $\beta_t = \{\beta_{t=1}^T\}$. We formulate the process as follows:

$$q(\mathbf{F}_m^{(1:T)} | \mathbf{F}_m^{(0)}) = \prod_{t=1}^T q(\mathbf{F}_m^{(t)} | \mathbf{F}_m^{(t-1)}), \quad (13)$$

$$q(\mathbf{F}_m^{(t)} | \mathbf{F}_m^{(t-1)}) = \mathcal{N}(\mathbf{F}_m^t; \sqrt{1 - \beta_t} \mathbf{F}_m^{(t-1)}, \beta_t \mathbf{I}), \quad (14)$$

For DDPM, the noising process is simplified as:

$$\mathbf{F}_m^{(t)} = \sqrt{\bar{\gamma}_t} \cdot \mathbf{F}_m^{(0)} + \sqrt{1 - \bar{\gamma}_t} \epsilon, \quad (15)$$

where $\bar{\gamma}_t = \bar{\gamma}_{t=1}^T$ denotes a fixed variance schedule that ensures the features follow a Gaussian distribution.

In the denoising stage, a modality-specific network $\mathcal{D}(\cdot)$ is designed to reverse the noisy distribution of each modality $\mathbf{F}_{m_c}^{(t)}$ and generate the clean target distribution $\tilde{\mathbf{F}}_m^{(0)}$. The conditional denoising process is formulated as:

$$p_\theta(\mathbf{F}_m^{(t-1)} | \mathbf{F}_m^{(t)}, \mathbf{F}_{m_c}^{(t)}) = \mathcal{N}(\mathbf{F}_m^{(t-1)}; \mu_\theta(\mathbf{F}_m^{(t)}, \mathbf{F}_{m_c}^{(t)}, t), \Sigma_\theta(\mathbf{F}_m^{(t)}, \mathbf{F}_{m_c}^{(t)}, t)), \quad (16)$$

where $\mathbf{F}_{m_c}^{(t)}$ denotes the conditional modality providing cross-modal guidance and $\mathbf{F}_m^{(t)}$ is the target modality being reconstructed. After that, the model minimizes both the mean squared error of noise prediction and the constraint loss of denoised features:

$$\mathcal{L}_{DM} = \mathbb{E}_{\mathbf{F}_m^{(0)}, c, \epsilon, t} [\|\epsilon - \epsilon_\theta(\mathbf{F}_m^{(t)}, t, c)\|_2^2] + \|\tilde{\mathbf{F}}_m^{(0)} - \mathbf{F}_m^{(0)}\|_2^2. \quad (17)$$

Finally, we propose a post-fusion strategy to fuse pseudo-features with the original ones. The reversed pseudo modality $\tilde{\mathbf{F}}_m$ is evaluated to obtain a new belief mass $\hat{\mathbf{b}}$, which collaborates with the original belief $\tilde{\mathbf{b}}^{(L)}$ to drive the post fusion. As Eq. (18), if the belief difference is below τ , the two modality features are averaged. Otherwise, the feature with the higher belief value is used:

$$\bar{\mathbf{F}}_m = \begin{cases} \frac{\mathbf{F}_m + \tilde{\mathbf{F}}_m}{2}, & \text{if } |\hat{\mathbf{b}} - \tilde{\mathbf{b}}^{(L)}| < \tau, \\ \mathbf{F}_{m, \max}, & \text{otherwise.} \end{cases} \quad (18)$$

where $\mathbf{F}_{m, \max}$ corresponds to the feature with the higher belief, and τ is the belief difference threshold.

Methods	RGBNT201				MSVR310		RGBNT100		WMVEID863			
	mAP	R@1	R@5	R@10	mAP	R@1	mAP	R@1	mAP	R@1	R@5	R@10
HAMNet [15]	27.7	26.3	41.5	51.7	27.1	42.3	74.5	93.3	45.6	48.5	63.1	68.8
PFNet [49]	38.5	38.9	52.0	58.4	23.5	37.4	68.1	94.1	50.1	55.9	68.7	75.1
IEEE [36]	46.4	47.1	58.5	64.2	21.0	41.0	61.3	87.8	45.9	48.6	64.3	67.9
CCNet [50]	-	-	-	-	36.4	55.2	77.2	96.3	50.3	52.7	69.6	75.1
TransReID* [6]	65.2	66.9	78.5	84.7	33.4	48.9	75.6	92.9	67.0	74.7	79.5	82.4
FACENet* [51]	-	-	-	-	36.2	54.1	81.5	96.9	69.8	77.0	81.0	84.2
EDITOR* [45]	66.5	68.3	81.1	88.2	39.0	49.3	82.1	96.4	65.6	73.8	80.0	82.3
RSCNet* [43]	68.2	72.5	-	-	39.5	49.6	82.3	96.6	-	-	-	-
TOP-ReID* [32]	72.3	76.6	84.7	89.4	35.9	44.6	81.2	96.4	67.7	75.3	80.8	83.5
ICPL-ReID* [18]	75.1	77.4	84.2	87.9	56.9	77.7	87.0	<u>98.6</u>	67.2	74.0	81.3	85.6
PromptMA* [48]	78.4	80.9	87.0	88.9	55.2	64.5	85.3	97.4	-	-	-	-
MambaPro* [33]	78.9	83.4	89.8	91.9	47.0	56.5	83.9	94.7	69.5	76.9	80.6	83.8
DeMo* [34]	79.0	82.3	88.8	92.0	49.2	59.8	86.2	97.6	68.8	77.2	81.5	83.8
IDEA* [35]	80.2	82.1	90.0	93.3	47.0	62.4	87.2	96.5	-	-	-	-
UGG-ReID* [30]	81.2	86.8	92.0	<u>94.7</u>	60.1	78.0	88.0	98.1	72.6	80.8	84.2	87.2
RoDI*	<u>84.1</u>	<u>87.2</u>	<u>92.0</u>	93.2	64.1	77.2	<u>88.5</u>	97.6	<u>73.9</u>	<u>81.9</u>	<u>86.5</u>	<u>87.7</u>
DINOV3 [†] [27]	81.0	83.3	91.0	93.4	68.2	<u>83.3</u>	87.0	98.5	70.1	80.8	84.3	87.7
IDEA [†] [35]	81.2	83.2	91.0	94.4	67.0	82.4	88.2	96.5	-	-	-	-
DeMo [†] [34]	82.8	85.4	91.1	93.1	<u>68.7</u>	81.6	88.4	96.1	72.5	79.9	85.2	89.2
RoDI[†]	85.3	87.9	93.0	94.8	71.8	84.8	89.0	99.1	74.2	82.2	86.7	89.7

Table 1. Comparison with the state-of-the-art methods on four benchmarks: RGBNT201, MSVR310, RGBNT100, and WMVEID863. Best results are in **bold**, the second bests are underlined. [†] denotes DINOv3-based methods, * indicates CLIP-based, * indicates ViT-based while others are CNN-based ones.

3.5. Optimization and Inference

Following prior work [6, 42], the model is trained with the classification loss $\mathcal{L}_{cls}$ and triplet loss $\mathcal{L}_{tri}$ based on identity labels Y . The total training loss is formulated as

$$\mathcal{L}_{final} = \mathcal{L}_{cls} + \mathcal{L}_{tri} + \lambda_1 \cdot \mathcal{L}_{KL} + \lambda_2 \cdot \mathcal{L}_{DM}, \quad (19)$$

where λ_1 and λ_2 are hyperparameters used to balance the contribution of the KL loss and the generative loss. In the inference stage, the fusion module is independent of the subjective opinion metrics and modality rolling.

4. Experiments

4.1. Datasets and Evaluation Protocols

Datasets. We evaluate our method on four public multi-modal benchmarks. RGBNT201 [49] contains 4,787 samples from 201 identities across three spectra (RGB, NIR, TIR) and four camera views. MSVR310 [50] consists of 2,087 triplets from 310 vehicles, captured under eight viewpoints with strong illumination, reflection, and color distortion variations. RGBNT100 [15] expands to 17,250 image triples of 100 vehicles with additional thermal modality, while WMVEID863 [51] provides 4,709 triplets from 863 vehicles under severe glare conditions, serving as the largest multi-modal benchmark.

Evaluation Protocols. We follow the standard evaluation protocol used in prior works [49–51], reporting mean Aver-

age Precision (mAP) and Cumulative Matching Characteristic (CMC) scores at Rank-K (K=1, 5, 10).

4.2. Implementation Details

All experiments are conducted on one NVIDIA RTX 4090 GPU using the PyTorch framework. We use DINO-v3 [27] ViT-B/16 (distilled) and CLIP [25] ViT-B/16 as visual encoders. Input images are resized to $224 \times 224 / 256 \times 128$ for person datasets and $224 \times 224 / 128 \times 256$ for vehicle datasets, depending on the backbone. The mini-batch size is set to 64, with 8 images per identity. Standard augmentations are combined, including random flipping, padding, erasing, and cropping [52]. Adam [13] optimizer is used with the learning rate of $3.5e-4$, weight decay of 0.0001, and momentum set to 0.9. The 10-epoch warmup is employed, linearly increasing the learning rate from $0.1 \times$ to the target value. Loss hyperparameters are set to $\lambda_1 = 1.0$ and $\lambda_2 = 0.1$ to keep training balance.

4.3. Comparison with State-of-the-Arts

Performance on Person Dataset. In Tab. 1, we first evaluate the DINOv3-based models on RGBNT201. The reproduced IDEA[†] and DeMo[†] improve upon the DINOv3[†] baseline, yet our RoDI[†] achieves the highest accuracy with 85.3% mAP and 87.9% Rank-1, surpassing IDEA[†] by 4.1% and 4.7% points respectively. These gains verify that RoDI effectively quantifies fusion mass and balances feature learning via evidence measure and rolling-

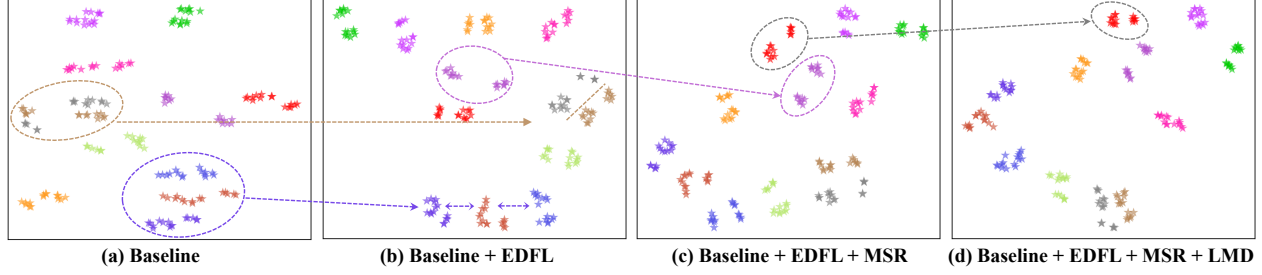


Figure 3. T-SNE [29] visualization of the feature distribution on RGBNT201 (a) Baseline, (b) Baseline + EDFL, (c) Baseline + EDFL + MSR, and (d) Baseline + EDFL + MSR + LMD (Ours).

Backbone	Index	Modules			Metrics			
		EDFL	MSR	LMD	mAP	R@1	R@5	R@10
CLIP	(a)	×	×	×	76.0	77.6	85.4	89.6
	(b)	✓	×	×	79.3	83.1	90.6	93.5
	(c)	✓	✓	×	82.7	85.9	90.7	93.5
	(d)	✓	✓	✓	84.1	87.2	92.0	93.2
DINOv3	(a)	×	×	×	81.0	83.3	91.0	93.4
	(b)	✓	×	×	82.3	84.2	90.4	93.1
	(c)	✓	✓	×	84.0	85.2	92.2	94.6
	(d)	✓	✓	✓	85.3	87.9	93.0	94.8

Table 2. Ablation studies of different components.

Index	MSR				Metrics			
	Rolling	Imbalance Mass	$\mathcal{L}_{KL}$		mAP	R@1	R@5	R@10
(a)	×	×	×		82.2	83.6	91.3	92.7
(b)	✓	×	×		83.4	86.6	92.3	94.5
(c)	✓	✓	×		84.4	86.8	92.5	94.2
(d)	✓	✓	✓		85.3	87.9	93.0	94.8

Table 3. Comparison with different components in MSR.

denoising optimization. For the CLIP-based setting, RoDI* also demonstrates clear superiority, reaching 84.1% mAP and 87.2% Rank-1, outperforming UGG-ReID* and IDEA* by large margins. The consistent improvements across both backbones highlight the generalization of RoDI and its ability to maintain fairness among modalities while enhancing discriminative representation quality.

Performance on Vehicle Dataset. For vehicle datasets, RoDI[†] consistently leads all DINOv3-based methods. It achieves 71.8% mAP and 84.8% Rank-1 on MSVR310, improving over DeMo[†] and IDEA[†] by up to 4.8% mAP and 2.4% Rank-1. On RGBNT100 and WMVeID863, RoDI[†] maintains the best results with 89.0% and 74.2% mAP, confirming its robustness across spectral variations. Under the CLIP backbone, RoDI* again outperforms UGG-ReID* and IDEA*, reaching 64.1% mAP and 77.2% Rank-1 on MSVR310. The consistent superiority across domains and backbones demonstrates that RoDI achieves balanced, conflict-free multi-modal fusion.

4.4. Ablation Studies

We leverage the pre-trained DINOv3 and CLIP encoder as encoders and extend them into a three-branch network as

Index	LMD		Metrics			
	Belief Mass	Post Fusion	mAP	R@1	R@5	R@10
(a)	×	×	83.1	85.7	91.8	94.5
(b)	✓	×	84.2	86.3	92.2	95.0
(c)	×	✓	83.9	86.0	92.2	95.2
(d)	✓	✓	85.3	87.9	93.0	94.8

Table 4. Comparison with different components in LMD.

our baseline for analysis on the RGBNT201.

Effects of Key Modules. Tab. 2 reports the ablation study of each component within RoDI. Starting from the plain baseline, introducing EDFL substantially improves mAP from 76.0% to 79.3% with CLIP and from 81.0% to 82.3% with DINOv3, validating the benefit of evidence-based fusion measurement. Adding MSR further boosts the CLIP model to 82.7% mAP and the DINOv3 model to 84.0% mAP, confirming that rolling-based query alignment effectively restores the balance of fusion. Finally, integrating LMD yields the full RoDI, which attains 84.1%/85.3% mAP and 87.2%/87.9% Rank-1 on CLIP and DINOv3 backbones, respectively. These results indicate that each module effectively enhances fusion balance and robustness.

Effects of Modality Rolling. Tab. 3 presents the analysis of different components in MSR. Tab. 3 (a) serves as the baseline, achieving 82.2% mAP and 83.6% Rank-1. Tab. 3 (b) introduces Rolling, which improves modality alignment and raises performance to 83.4% mAP and 86.6% Rank-1. Tab. 3 (c) adds Imbalance Mass, further enhancing balance and improving results to 84.4% mAP and 86.8% Rank-1. Tab. 3 (d) incorporates the KL regularization, keeping cross-modal consistency, and reaching 85.3% mAP and 87.9% Rank-1. These incremental gains demonstrate the effectiveness of each component in MSR.

Effects of Modality Denoising. Tab. 4 shows the performance of different components in LMD. Tab. 4 (a) shows the baseline, achieving 83.1% mAP and 85.7% Rank-1. When Belief Mass is introduced in Tab. 4 (b), the model achieves 84.2% mAP and 86.3% Rank-1, showing that belief mass guided token selection enhances local reliability. In Tab. 4 (c), adding Post Fusion instead improves to 83.9% mAP and 86.0% Rank-1, validating the benefit of denoised

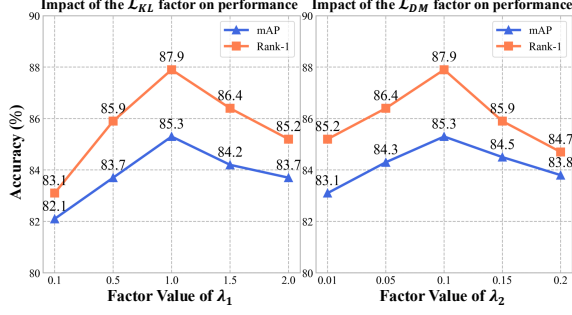


Figure 4. Performance comparison with different loss factors.

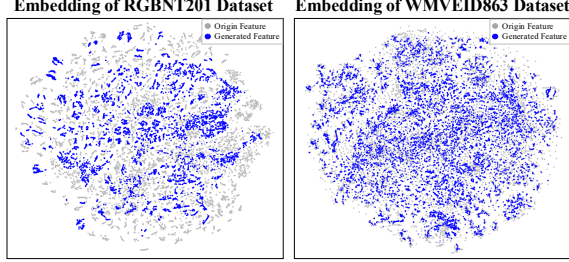


Figure 5. Visualization of the feature embeddings.

feature reconstruction. Finally, the complete LMD configuration in Tab. 4 (d) reaches 85.3% mAP and 87.9% Rank-1, outperforming all variants. These results validate the effectiveness of components in LMD.

Effects of Loss Factor. Fig. 4 shows the model achieves the best results when $\lambda_1 = 1.0$ and $\lambda_2 = 0.1$, reaching 85.3% mAP and 87.9% Rank-1. Too small weights weaken rolling and denoising, while overly large ones cause over-constraint and overfitting. Both curves remain stable around these values, indicating that moderate loss scaling yields the most balanced and robust fusion performance.

4.5. Visualization Analysis

Feature Distribution. Fig. 3 visualizes the feature distributions of different components. Comparing Fig. 3 (a) and (b), the baseline shows scattered and overlapping clusters, while EDFL encourages intra-class compactness. In Fig. 3 (c), MSR further balances fusion and enhances inter-class separation. Finally, in Fig. 3 (d), LMD suppresses noisy modality conflicts, yielding cleaner and more compact clusters. These results validate that our proposed modules make the learned features more discriminative and well-separated.

Modality Weight Distribution in MSR. To give deeper insights, Fig. 6 shows the fusion weights estimated by EDFL. The left plot in Fig. 6, without MSR, the fusion distribution exhibits several noticeable secondary peaks with weights of 0.140, 0.190, and 0.173, indicating modality suppression where certain modalities are insufficiently utilized. Within the main peak, the TIR modality shows a higher average weight of 0.373 compared to RGB and NIR, revealing its

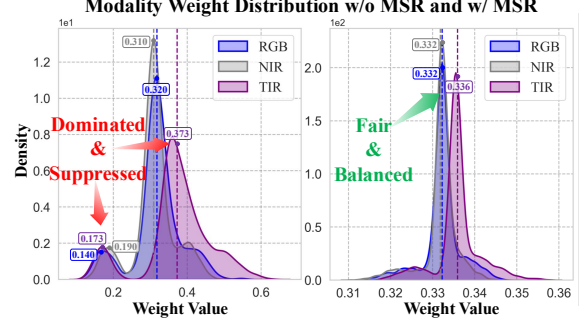


Figure 6. Analysis of modality weight distribution.

dominant influence in the fusion process. After introducing MSR, the right plot in Fig. 6 shows that the fusion weights become significantly more balanced, the dominance of the TIR modality diminishes, and the secondary peaks disappear. These observations confirm that MSR effectively equalizes modality contributions and prevents dominance from a single modality during fusion.

Embedding Visualization in LMD. To intuitively verify the superiority of the proposed LMD, the generated features on RGBNT201 and WMVEID863 are visualized in Fig. 5 using t-SNE [29]. The gray points represent the original features, while the blue points denote those generated by LMD. The high overlap between the two distributions indicates strong semantic consistency, confirming the generated features can serve as reliable substitutes for their original counterparts. In addition, the original features exhibit a more scattered distribution with ambiguous outliers, whereas the generated ones are more compact. These results validate the effectiveness of the denoising process in reducing noise and enhancing feature reliability.

5. Conclusion

In this paper, we revisit multi-modal object ReID from the perspectives of fusion evaluation and optimization. We propose RoDI (Rolling and Denoising), a unified framework that jointly addresses fusion imbalance and fusion conflict. Through the Evidence-aware Dynamic Fusion Learning (EDFL) module, RoDI quantifies belief reliability and uncertainty during fusion, offering interpretable measures of fusion quality. The Modality Synergistic Rolling (MSR) module aligns inter-modal evidence distributions by estimating balanced anchors and rotating fusion queries to recover equitable feature collaboration. The Local Modality Denoising (LMD) module further refines local tokens via diffusion-based reconstruction, generating high-quality pseudo-modal features that suppress conflicts from degraded modalities. Extensive experiments across four public benchmarks verify that RoDI achieves state-of-the-art results and reveals a new direction toward a measurable, balanced, and conflict-free fusion framework.

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References

- [1] Jie Chang, Zhonghao Lan, Changmao Cheng, and Yichen Wei. Data uncertainty learning in face recognition. In *CVPR*, 2020. 5
- [2] Abhra Chaudhuri, Anjan Dutta, Tu Bui, and Serban Georgescu. A closer look at multimodal representation collapse. In *ICML*, 2025. 2, 3, 4
- [3] Wei-Ting Chen, I-Hsiang Chen, Chih-Yuan Yeh, Hao-Hsiang Yang, Jian-Jiun Ding, and Sy-Yen Kuo. Sjd-vehicle: Semi-supervised joint defogging learning for foggy vehicle re-identification. In *AAAI*, 2022. 1
- [4] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *ICLR*, 2021. 4
- [5] Zongbo Han, Changqing Zhang, Huazhu Fu, and Joey Tianyi Zhou. Trusted multi-view classification with dynamic evidential fusion. *IEEE TPAMI*, 2022. 3
- [6] Shuting He, Hao Luo, Pichao Wang, Fan Wang, Hao Li, and Wei Jiang. Transreid: Transformer-based object re-identification. In *ICCV*, 2021. 6
- [7] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *NeurIPS*, 2020. 5
- [8] Haojian Huang, Chuanyu Qin, Zhe Liu, Kaijing Ma, Jin Chen, Han Fang, Chao Ban, Hao Sun, and Zhongjiang He. Trusted unified feature-neighborhood dynamics for multi-view classification. In *AAAI*, 2025. 2, 3
- [9] Yu Huang, Junyang Lin, Chang Zhou, Hongxia Yang, and Longbo Huang. Modality competition: What makes joint training of multi-modal network fail in deep learning? (Provably). In *ICML*, 2022. 2, 4
- [10] Sergey Ioffe and Christian Szegedy. Batch normalization: Accelerating deep network training by reducing internal covariate shift. In *ICML*, 2015. 4
- [11] Norman Lloyd Johnson, Samuel Kotz, and Narayanaswamy Balakrishnan. *Continuous multivariate distributions*. Wiley New York, 1972. 4
- [12] Audun Jøsang. *Subjective logic*. Springer, 2016. 4
- [13] Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014. 6
- [14] Qingming Leng, Mang Ye, and Qi Tian. A survey of open-world person re-identification. *IEEE TCSVT*, 2020. 1
- [15] Hongchao Li, Chenglong Li, Xianpeng Zhu, Aihua Zheng, and Bin Luo. Multi-spectral vehicle re-identification: A challenge. In *AAAI*, 2020. 6
- [16] Shenshen Li, Chen He, Xing Xu, Fumin Shen, Yang Yang, and Heng Tao Shen. Adaptive uncertainty-based learning for text-based person retrieval. In *AAAI*, 2024. 2, 3
- [17] Shihao Li, Aihua Zheng, Andong Lu, Jin Tang, and Jixin Ma. Next: Multi-grained mixture of experts via text-modulation for multi-modal object re-identification. *arXiv preprint arXiv:2505.20001*, 2025. 2
- [18] Shihao Li, Chenglong Li, Aihua Zheng, Jin Tang, and Bin Luo. Icpl-reid: Identity-conditional prompt learning for multi-spectral object re-identification. *IEEE TMM*, 2026. 6, 2
- [19] Shihao Li, Chenglong Li, Aihua Zheng, Jin Tang, and Bin Luo. Deep: Decoupled semantic prompt learning, guiding and embedding for multi-spectral object re-identification. *IEEE TMM*, 2026. 2
- [20] Andong Lu, Zhang Zhang, Yan Huang, Yifan Zhang, Chenglong Li, Jin Tang, and Liang Wang. Illumination distillation framework for nighttime person re-identification and a new benchmark. *IEEE TMM*, pages 1–14, 2023. 1
- [21] Hao Luo, Youzhi Gu, Xingyu Liao, Shenqi Lai, and Wei Jiang. Bag of tricks and a strong baseline for deep person re-identification. In *CVPRW*, 2019. 1
- [22] Yujie Mo, Yajie Lei, Jialie Shen, Xiaoshuang Shi, Heng Tao Shen, and Xiaofeng Zhu. Disentangled multiplex graph representation learning. In *ICML*, 2023. 5
- [23] Haotian Ni, Yake Wei, Hang Liu, Gong Chen, Chong Peng, Hao Lin, and Di Hu. Rollingq: Reviving the cooperation dynamics in multimodal transformer. In *ICML*, 2025. 2, 3
- [24] Hossein R. Nowdeh, Jie Ji, Xiaolong Ma, and Fatemeh Afghah. Modality-aware SAM: Sharpness-aware-minimization driven gradient modulation for harmonized multimodal learning. In *NeurIPS*, 2025. 2, 3
- [25] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *ICML*, 2021. 6
- [26] Murat Sensoy, Lance Kaplan, and Melih Kandemir. Evidential deep learning to quantify classification uncertainty. In *NeurIPS*, 2018. 4
- [27] Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, Cijo Jose, Vasil Khalidov, Marc Szafraniec, Seungeun Yi, Michaël Ramamonjisoa, Francisco Massa, Daniel Haziza, Luca Wehrstedt, Jianyuan Wang, Timothée Darcet, Théo Moutakanni, Leonel Sentana, Claire Roberts, Andrea Vedaldi, Jamie Tolan, John Brandt, Camille Couprie, Julien Mairal, Hervé Jégou, Patrick Labatut, and Piotr Bojanowski. DINOv3, 2025. 6, 2
- [28] Yifan Sun, Liang Zheng, Yi Yang, Qi Tian, and Shengjin Wang. Beyond part models: Person retrieval with refined part pooling (and A strong convolutional baseline). In *ECCV*, 2018. 1, 2
- [29] Laurens Van der Maaten and Geoffrey Hinton. Visualizing data using t-sne. *Journal of machine learning research*, 9 (11), 2008. 7, 8
- [30] Xixi Wan, Aihua Zheng, Bo Jiang, Beibei Wang, Chenglong Li, and Jin Tang. Ugg-reid: Uncertainty-guided graph model for multi-modal object re-identification. In *NeurIPS*, 2025. 1, 2, 6

- [31] Xixi Wan, Aihua Zheng, Zi Wang, Bo Jiang, Jin Tang, and Jixin Ma. Reliable multi-modal object re-identification via modality-aware graph reasoning. *arXiv preprint arXiv:2504.14847*, 2025. 1
- [32] Yuhao Wang, Xuehu Liu, Pingping Zhang, Hu Lu, Zhengzheng Tu, and Huchuan Lu. Top-reid: Multi-spectral object re-identification with token permutation. In *AAAI*, 2024. 1, 2, 4, 6
- [33] Yuhao Wang, Xuehu Liu, Tianyu Yan, Yang Liu, Aihua Zheng, Pingping Zhang, and Huchuan Lu. Mambapro: Multi-modal object re-identification with mamba aggregation and synergistic prompt. In *AAAI*, 2025. 4, 6, 2
- [34] Yuhao Wang, Yang Liu, Aihua Zheng, and Pingping Zhang. Demo: Decoupled feature-based mixture of experts for multi-modal object re-identification. In *AAAI*, 2025. 1, 2, 6
- [35] Yuhao Wang, Yongfeng Lv, Pingping Zhang, and Huchuan Lu. IDEA: inverted text with cooperative deformable aggregation for multi-modal object re-identification. In *CVPR*, 2025. 2, 6
- [36] Zi Wang, Chenglong Li, Aihua Zheng, Ran He, and Jin Tang. Interact, embed, and enlarge: Boosting modality-specific representations for multi-modal person re-identification. In *AAAI*, 2022. 6
- [37] Yake Wei, Di Hu, Henghui Du, and Ji-Rong Wen. On-the-fly modulation for balanced multimodal learning. *IEEE TPAMI*, 2024. 2, 3
- [38] Yake Wei, Siwei Li, Ruoxuan Feng, and Di Hu. Diagnosing and re-learning for balanced multimodal learning. In *ECCV*, 2024. 2
- [39] Nan Wu, Stanislaw Kamil Jastrzebski, Kyunghyun Cho, and Krzysztof J. Geras. Recognizing and overcoming the greedy nature of learning in multi-modal deep neural networks. In *ICLR*, 2022. 2, 3, 4
- [40] Cai Xu, Jiajun Si, Ziyu Guan, Wei Zhao, Yue Wu, and Xiyue Gao. Reliable conflictive multi-view learning. In *AAAI*, 2024. 2, 3
- [41] Xi Yang, Wenjiao Dong, De Cheng, Nannan Wang, and Xinbo Gao. Tienet: A tri-interaction enhancement network for multimodal person reidentification. *IEEE TNNLS*, 2025. 1
- [42] Mang Ye, Jianbing Shen, Gaojie Lin, Tao Xiang, Ling Shao, and Steven CH Hoi. Deep learning for person re-identification: A survey and outlook. *IEEE TPAMI*, 2021. 1, 6
- [43] Zhi Yu, Zhiyong Huang, Mingyang Hou, Jiaming Pei, Yan Yan, Yushi Liu, and Daming Sun. Representation selective coupling via token sparsification for multi-spectral object re-identification. *IEEE TCSVT*, 2024. 6
- [44] Quanxing Zha, Xin Liu, Yiu-ming Cheung, Xing Xu, Nannan Wang, and Jianjia Cao. UGNCL: uncertainty-guided noisy correspondence learning for efficient cross-modal matching. In *ACM MM*, 2024. 2
- [45] Pingping Zhang, Yuhao Wang, Yang Liu, Zhengzheng Tu, and Huchuan Lu. Magic tokens: Select diverse tokens for multi-modal object re-identification. In *CVPR*, 2024. 1, 2, 6
- [46] Qingyang Zhang, Haitao Wu, Changqing Zhang, Qinghua Hu, Huazhu Fu, Joey Tianyi Zhou, and Xi Peng. Provable dynamic fusion for low-quality multimodal data. In *ICML*, 2023. 3
- [47] Qingyang Zhang, Yake Wei, Zongbo Han, Huazhu Fu, Xi Peng, Cheng Deng, Qinghua Hu, Cai Xu, Jie Wen, Di Hu, et al. Multimodal fusion on low-quality data: A comprehensive survey. *arXiv preprint arXiv:2404.18947*, 2024. 3
- [48] Shizhou Zhang, Wenlong Luo, De Cheng, Yinghui Xing, Guoqiang Liang, Peng Wang, and Yanning Zhang. Prompt-based modality alignment for effective multi-modal object re-identification. *IEEE TIP*, 2025. 1, 2, 4, 6
- [49] Aihua Zheng, Zi Wang, Zi-Han Chen, Chenglong Li, and Jin Tang. Robust multi-modality person re-identification. In *AAAI*, 2021. 1, 2, 6
- [50] Aihua Zheng, Xianpeng Zhu, Zhiqi Ma, Chenglong Li, Jin Tang, and Jixin Ma. Cross-directional consistency network with adaptive layer normalization for multi-spectral vehicle re-identification and a high-quality benchmark. *INFFUS*, 2023. 6
- [51] Aihua Zheng, Zhiqi Ma, Yongqi Sun, Zi Wang, Chenglong Li, and Jin Tang. Flare-aware cross-modal enhancement network for multi-spectral vehicle re-identification. *INFFUS*, 2025. 1, 2, 6
- [52] Zhun Zhong, Liang Zheng, Guoliang Kang, Shaozi Li, and Yi Yang. Random erasing data augmentation. In *AAAI*, 2020. 6

Rolling and Denoising: Rethinking Dynamic Modal Fusion for Multi-Modal Object Re-Identification

Supplementary Material

6. Introduction

In this supplementary material, we provide more comprehensive experiments and in-depth analyses to further investigate fusion imbalance and conflict denoising. Specifically, we address the following questions:

Sec.A How robust are the denoising network and fusion module under modality conflicts?

- Robustness analysis under Gaussian noise perturbations.
- Experiments on missing-modality scenarios.

Sec.B How does modality imbalance evolve during training?

- Analysis of modality weights in training.

Sec.C How is the computational efficiency of the model and the selection of hyperparams?

- Analysis of computational efficiency.
- Analysis of the TOP- K value in LMD.
- Analysis of diffusion steps used in LMD.

These analyses offer a deeper insight of our RoDI and further demonstrate the effectiveness of its core modules.

7. How robust are the denoising network and fusion module under modality conflicts?

7.1. Robustness analysis under Gaussian noise perturbations.

Fig. 7 demonstrates that RoDI[†] achieves the strongest resilience against progressive Gaussian corruption. With the DINOv3 backbone, RoDI[†] hold leading performance from $\varepsilon = 0$ to $\varepsilon = 30$, maintaining 61.5% mAP and 66.6% Rank-1, while DeMo[†] degrades more rapidly across both metrics. CLIP-based models such as UGG-ReID and DeMo drop off sharply, and ViT-based EDITOR breaks down even at moderate noise levels. However, compare with these methods, RoDI using the same CLIP backbone, remains consistently robust in low-quality scenarios. These results emphasize the effectiveness of our proposed denoising mechanism which not only counters noise-induced fusion conflicts but also enables the model to effectively fuse both easy and hard samples.

7.2. Experiments on missing-modality scenarios.

Tab. 5 evaluates the robustness of different methods under various missing-modality settings on RGBNT201. Among CLIP-based approaches, RoDI achieves the best average performance with 58.3% mAP and 59.2% Rank-1, surpassing PromptMA by 3.7% mAP and 4.7% Rank-1, and clearly

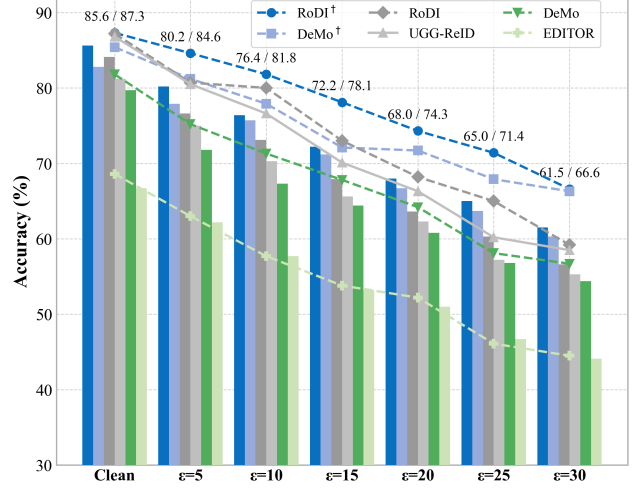


Figure 7. Robustness analysis on RGBNT201. Evaluation results with different levels of Gaussian noise added during testing after training on clean data. ε is the variance of Gaussian.

outperforming DeMo and TOP-ReID across most configurations. In single-modality cases, RoDI attains 68.3% mAP and 68.5% Rank-1 with only RGB available, 77.9% mAP and 80.5% Rank-1 with only NIR, and 62.9% mAP and 64.5% Rank-1 with only TIR, demonstrating strong capacity to exploit each individual spectrum. When two modalities are missing, RoDI shows particularly large gains in challenging settings such as “M (RGB+TIR)” and “M (NIR+TIR)”, where it significantly outperforms all baselines. In the DINOv3-based comparison, RoDI[†] achieves the highest average performance with 61.7% mAP and 64.0% Rank-1, improving over DeMo[†] by 4.5% mAP and 7.0% Rank-1, and dominating in five out of six missing-modality configurations. These results demonstrate that RoDI’s denoising module is capable of generating high-quality and reliable pseudo-modal representations, while its fusion module maintains strong robustness even under severe modality dropout.

8. How does modality imbalance evolve during training?

8.1. Analysis of modality weights in training.

Fig. 8 visualizes the evolution of modality balance weights estimated by EDFL during training. Each fusion query is initialized from its own modality and thus exhibits a

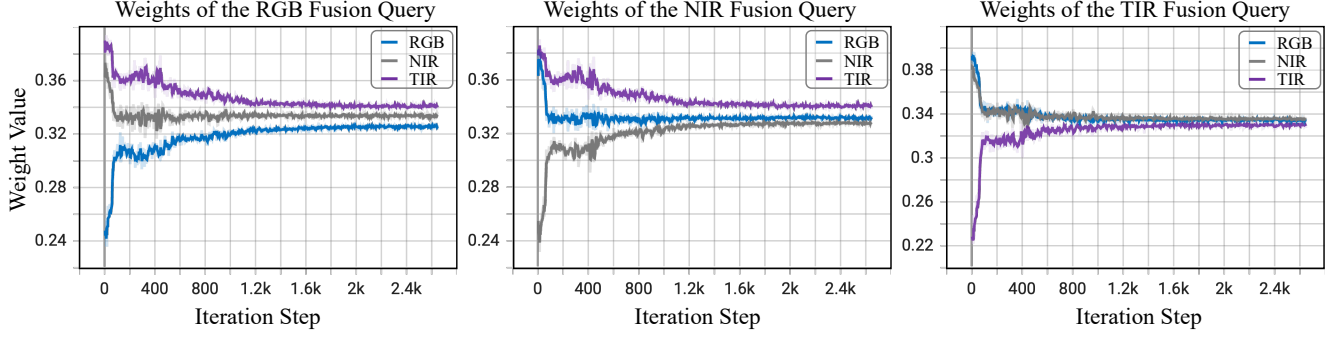


Figure 8. The evolution of modality balance weights estimated by EDFL during training.

Methods	M (RGB)		M (NIR)		M (TIR)		M (RGB+NIR)		M (RGB+TIR)		M (NIR+TIR)		Avg.	
	mAP	R@1	mAP	R@1	mAP	R@1	mAP	R@1	mAP	R@1	mAP	R@1	mAP	R@1
PCB [28]	23.6	24.2	24.4	25.1	19.9	14.7	20.6	23.6	11.0	6.8	18.6	14.4	19.7	18.1
TOP-ReID [32]	54.4	57.5	64.3	67.6	51.9	54.5	35.3	35.4	26.2	26.0	34.1	31.7	44.4	45.4
DeMo [34]	63.3	65.3	<u>72.6</u>	<u>75.7</u>	56.2	54.1	45.6	46.5	26.3	24.9	40.3	38.5	50.7	50.8
PromptMA [48]	67.4	68.4	72.5	75.7	58.9	57.3	51.5	53.0	33.3	30.4	43.9	42.2	54.6	54.5
RoDI	68.3	68.5	77.9	80.5	62.9	64.5	47.7	49.0	44.1	43.3	48.6	49.6	58.3	59.2
DeMo [†] [34]	71.3	71.8	<u>77.2</u>	78.8	<u>63.0</u>	61.7	56.5	57.1	<u>32.4</u>	<u>32.1</u>	42.9	40.7	<u>57.2</u>	<u>57.0</u>
RoDI[†]	<u>69.0</u>	<u>70.1</u>	80.0	82.7	66.4	69.5	<u>52.3</u>	<u>53.5</u>	49.0	50.4	53.2	57.7	61.7	64.0

Table 5. Performance of the missing-modalities settings on RGBNT201. “M (X)” means missing the X spectral modality. Best results are in **bold**, the second bests are underlined.

strong bias against the other two modalities at the early stage. As training progresses, the model gradually learns to redistribute attention and incorporate information from multiple spectra. However, the upper trajectories reveal that fusion queries still exhibit noticeable bias toward the two non-origin modalities before convergence, and this bias diminishes only as the modality fusion process becomes more mature. These observations confirm that EDFL effectively exposes the dynamic imbalance across modalities and highlights the necessity of subsequent rolling-based re-balancing.

9. How is the computational efficiency of the model and the selection of hyperparams?

9.1. Analysis of computational efficiency.

Tab. 6 reports a comprehensive comparison of computational efficiency across state-of-the-art methods. Under the CLIP backbone, RoDI attains the best overall performance 84.1%/87.2% mAP/Rank-1 while using only 90.6M parameters and 38.7G FLOPs, which significantly lighter than methods such as UGG-ReID (103.2 M), PromptMA (107.9 M), and TOP-ReID (324.5 M). Even compared with lighter models such as ICPL-ReID, RoDI provides over +9.0% mAP improvement with similar computational cost. Under the DINOv3 backbone, RoDI[†] exhibits the same advantage,

Methods	Params	FLOPs	Throughput	Metrics	
	(M)	(G)	(TPS)	mAP	R@1
EDITOR [45]	119.3	40.8	335.1	66.5	68.3
TOP-ReID [32]	324.5	35.5	398.9	72.3	76.6
ICPL-ReID [18]	45.4	39.8	358.4	75.1	77.4
PromptMA [48]	107.9	36.2	343.5	78.4	80.9
DeMo [34]	98.8	35.1	403.6	79.0	82.3
MambaPro [33]	74.8	52.4	243.2	78.9	83.4
IDEA [35]	91.7	43.7	299.5	80.2	82.1
UGG-ReID [30]	103.2	35.0	371.4	81.2	86.8
RoDI	90.6	38.7	299.6	84.1	87.2
DINOv3 [†] [27]	86.1	54.0	238.6	81.0	83.3
IDEA [†] [35]	94.0	63.5	189.6	81.2	83.2
DeMo [†] [34]	113.4	54.5	234.8	82.8	85.4
RoDI[†]	95.1	64.0	191.1	85.3	87.9

Table 6. Comparison with state-of-the-art methods on the size of learnable parameters, flops, and throughput on RGBNT201.

outperforming all DINOv3-based methods while remaining smaller than DeMo[†]. These results confirm that RoDI provides an optimal balance between efficiency and performance, delivering state-of-the-art accuracy without incurring additional computational overhead.

9.2. Analysis of the TOP-K value in LMD.

As illustrated in Fig. 9, the performance of LMD varies with different TOP-K selections guided by the belief scores es-

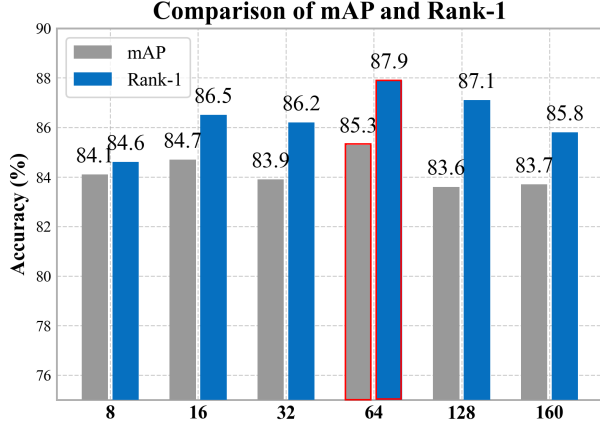


Figure 9. Performance comparison with different TOP- K value.

Denoising Steps	mAP	R@1
1	78.8	80.3
5	81.7	84.3
10	83.8	86.2
20	85.3	87.9
30	<u>83.8</u>	89.0
40	76.0	84.7
50	63.1	74.6

Table 7. Performance under different denoising step settings on RGBNT201 using DINOv3-Base.

timated in EDFL. Moderate K values (16 and 32) improve Rank-1 accuracy compared to very small selections, while $K=64$ provides the best overall results, achieving 85.3% mAP and 87.9% Rank-1. This indicates that selecting a sufficiently rich set of high-belief local tokens is crucial for stable fusion learning. When K becomes excessively large (128 or 160), performance drops, suggesting that including too many low-confidence regions dilutes the reliability of the fused evidence. These results confirm that evidence-based selection of local features effectively discard harmful regions and substantially mitigates fusion conflicts caused by noisy or low-quality local tokens.

9.3. Analysis of denoising steps used in LMD.

As shown in Tab. 7, the model benefits from increasing denoising steps up to 20, where it achieves the highest results 85.3%/87.9% mAP/Rank-1. Fewer steps (1–10) lead to under-denoising, while more steps (40–50) cause severe performance degradation due to over-smoothing and information loss. Although Rank-1 peaks at 30 steps, the drop in mAP indicates less reliable retrieval overall. These trends suggest that a moderate diffusion depth, such as 20 steps, yields the most stable and discriminative pseudo-modal representations.